

Task 1. Microservices Basics

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[Github repo](#)

1. Базова коректність та функціональність:

Приклади основних запитів та відповідей на них:

- POST /transaction:

The image displays three sequential screenshots of a REST client interface, showing the process of sending POST requests to the endpoint `http://localhost:3000/transaction` and receiving JSON responses.

First Screenshot: The request body is `{ "user_Id": "Alice", "amount": 5 }`. The response status is 200 OK, and the response body is `{ "transaction_ID": "4c485251-eea3-4837-a22d-0dbbfd2e14b2", "balance": 5 }`.

Second Screenshot: The request body is `{ "user_Id": "Bob", "amount": 15 }`. The response status is 200 OK, and the response body is `{ "transaction_ID": "2dbe6b0c-76cc-425d-88ec-a14390679b1a", "balance": 15 }`.

Third Screenshot: The request body is `{ "user_Id": "Alice", "amount": -20 }`. The response status is 200 OK, and the response body is `{ "transaction_ID": "eb8f2169-042e-4d31-9005-8bc82e54da8f", "balance": -15 }`.

- GET /user/:userId:

GET http://localhost:3000/user/Alice

Status: 200 OK Time: 30 ms Size: 466 B

```
1 {
2   "balance": -15,
3   "logs": [
4     {
5       "transaction_ID": "4c485251-eea3-4837-a22d-0dbbfd2e14b2",
6       "timestamp": "2026-02-22T11:21:03.058Z",
7       "msg": {
8         "user_Id": "Alice",
9         "amount": 5
10      }
11    },
12    {
13      "transaction_ID": "eb8f2169-042e-4d31-9005-8bc82e54da8f",
14      "timestamp": "2026-02-22T11:22:00.246Z",
15      "msg": {
16        "user_Id": "Alice",
17        "amount": -20
18      }
19    }
20  ]
21 }
```

- GET /users:

GET http://localhost:3000/accounts

Status: 200 OK Time: 8 ms Size: 206 B

```
1 {
2   "accounts": {
3     "Alice": -15,
4     "Bob": 15
5   }
6 }
```

Вигновігні логи у сервісах:

- facade-service:

```
{
  "level": 30, "time": 1771759263057, "pid": 18, "hostname": "1ecf83febd38", "reqId": "req-5", "req": {
    "method": "POST", "url": "/transaction", "host": "localhost:3000", "remoteAddress": "172.19.0.1", "remotePort": 34724
  }, "msg": "incoming request"
}, {
  "level": 30, "time": 1771759263063, "pid": 18, "hostname": "1ecf83febd38", "reqId": "req-5", "res": {
    "statusCode": 200, "responseTime": 5.860341999679804
  }, "msg": "request completed"
}, {
  "level": 30, "time": 1771759294121, "pid": 18, "hostname": "1ecf83febd38", "reqId": "req-6", "req": {
    "method": "POST", "url": "/transaction", "host": "localhost:3000", "remoteAddress": "172.19.0.1", "remotePort": 34724
  }, "msg": "incoming request"
}, {
  "level": 30, "time": 1771759294126, "pid": 18, "hostname": "1ecf83febd38", "reqId": "req-6", "res": {
    "statusCode": 200, "responseTime": 4.387272999621928
  }, "msg": "request completed"
}, {
  "level": 30, "time": 1771759320246, "pid": 18, "hostname": "1ecf83febd38", "reqId": "req-7", "req": {
    "method": "POST", "url": "/transaction", "host": "localhost:3000", "remoteAddress": "172.19.0.1", "remotePort": 34724
  }, "msg": "incoming request"
}, {
  "level": 30, "time": 1771759320250, "pid": 18, "hostname": "1ecf83febd38", "reqId": "req-7", "res": {
    "statusCode": 200, "responseTime": 4.362084999680519
  }, "msg": "request completed"
}, {
  "level": 30, "time": 1771759418127, "pid": 18, "hostname": "1ecf83febd38", "reqId": "req-8", "req": {
    "method": "GET", "url": "/user/Alice", "host": "localhost:3000", "remoteAddress": "172.19.0.1", "remotePort": 54552
  }, "msg": "incoming request"
}, {
  "level": 30, "time": 1771759418133, "pid": 18, "hostname": "1ecf83febd38", "reqId": "req-8", "res": {
    "statusCode": 200, "responseTime": 5.148719000630081
  }, "msg": "request completed"
}, {
  "level": 30, "time": 1771759571299, "pid": 18, "hostname": "1ecf83febd38", "reqId": "req-9", "req": {
    "method": "GET", "url": "/users", "host": "localhost:3000", "remoteAddress": "172.19.0.1", "remotePort": 53032
  }, "msg": "incoming request"
}, {
  "level": 30, "time": 1771759571299, "pid": 18, "hostname": "1ecf83febd38", "reqId": "req-9", "msg": "Route GET:/users not found"
}, {
  "level": 30, "time": 1771759571300, "pid": 18, "hostname": "1ecf83febd38", "reqId": "req-9", "res": {
    "statusCode": 404, "responseTime": 0.527222000624657
  }, "msg": "request completed"
}, {
  "level": 30, "time": 1771759593572, "pid": 18, "hostname": "1ecf83febd38", "reqId": "req-a", "req": {
    "method": "GET", "url": "/accounts", "host": "localhost:3000", "remoteAddress": "172.19.0.1", "remotePort": 53032
  }, "msg": "incoming request"
}, {
  "level": 30, "time": 1771759593577, "pid": 18, "hostname": "1ecf83febd38", "reqId": "req-a", "res": {
    "statusCode": 200, "responseTime": 5.335819000378251
  }, "msg": "request completed"
}
```

- logging-service:

```
{"level":30,"time":1771759228976,"pid":18,"hostname":"b02f96033f02","reqId":"req-6","res":{"statusCode":200},"responseTime":0.7388319997116923,"msg":"request completed"}
{"level":30,"time":1771759263061,"pid":18,"hostname":"b02f96033f02","reqId":"req-7","req":{"method":"POST","url":"/transaction","host":"logging-service:3001","remoteAddress":"172.19.0.4","remotePort":39992},"msg":"incoming request"}
{"level":30,"time":1771759294124,"pid":18,"hostname":"b02f96033f02","msg":"Logged transaction 4c485251-eea3-4837-a22d-0dbbfd2e14b2 for user Alice with amount 5 at 2026-02-22T11:21:03.058Z"}
{"level":30,"time":1771759263062,"pid":18,"hostname":"b02f96033f02","reqId":"req-7","res":{"statusCode":200},"responseTime":0.7498309994116426,"msg":"request completed"}
{"level":30,"time":1771759294124,"pid":18,"hostname":"b02f96033f02","reqId":"req-8","req":{"method":"POST","url":"/transaction","host":"logging-service:3001","remoteAddress":"172.19.0.4","remotePort":39992},"msg":"incoming request"}
{"level":30,"time":1771759294124,"pid":18,"hostname":"b02f96033f02","msg":"Logged transaction 2db6b0c-76cc-425d-88ec-a14390679b1a for user Bob with amount 15 at 2026-02-22T11:21:34.122Z"}
{"level":30,"time":1771759294125,"pid":18,"hostname":"b02f96033f02","reqId":"req-8","res":{"statusCode":200},"responseTime":0.6609259992837906,"msg":"request completed"}
{"level":30,"time":1771759320248,"pid":18,"hostname":"b02f96033f02","reqId":"req-9","req":{"method":"POST","url":"/transaction","host":"logging-service:3001","remoteAddress":"172.19.0.4","remotePort":39992},"msg":"incoming request"}
{"level":30,"time":1771759320248,"pid":18,"hostname":"b02f96033f02","msg":"Logged transaction eb8f2169-042e-4d31-9005-8bc82e54da8f for user Alice with amount -20 at 2026-02-22T11:22:00.246Z"}
{"level":30,"time":1771759320249,"pid":18,"hostname":"b02f96033f02","reqId":"req-9","res":{"statusCode":200},"responseTime":0.7762329997494817,"msg":"request completed"}
{"level":30,"time":1771759418130,"pid":18,"hostname":"b02f96033f02","reqId":"req-a","req":{"method":"GET","url":"/transaction/Alice","host":"logging-service:3001","remoteAddress":"172.19.0.4","remotePort":56994},"msg":"incoming request"}
{"level":30,"time":1771759418130,"pid":18,"hostname":"b02f96033f02","msg":"Returning transactions for user Alice, count: 2"}
{"level":30,"time":1771759418132,"pid":18,"hostname":"b02f96033f02","reqId":"req-a","res":{"statusCode":200},"responseTime":1.443262000568211,"msg":"request completed"}
```

- counter-service:

```
{"level":30,"time":1771759263062,"pid":18,"hostname":"f86c2c788f60","msg":"Updated balance for user Alice: 0 -> 5"}
{"level":30,"time":1771759263062,"pid":18,"hostname":"f86c2c788f60","reqId":"req-6","res":{"statusCode":200},"responseTime":0.5397230004891753,"msg":"request completed"}
{"level":30,"time":1771759294124,"pid":18,"hostname":"f86c2c788f60","reqId":"req-7","req":{"method":"POST","url":"/transaction","host":"counter-service:3002","remoteAddress":"172.19.0.4","remotePort":38686},"msg":"incoming request"}
{"level":30,"time":1771759294124,"pid":18,"hostname":"f86c2c788f60","msg":"Updated balance for user Bob: 0 -> 15"}
{"level":30,"time":1771759294125,"pid":18,"hostname":"f86c2c788f60","reqId":"req-7","res":{"statusCode":200},"responseTime":0.660525999632835,"msg":"request completed"}
{"level":30,"time":1771759320248,"pid":18,"hostname":"f86c2c788f60","reqId":"req-8","req":{"method":"POST","url":"/transaction","host":"counter-service:3002","remoteAddress":"172.19.0.4","remotePort":38686},"msg":"incoming request"}
{"level":30,"time":1771759320248,"pid":18,"hostname":"f86c2c788f60","msg":"Updated balance for user Alice: 5 -> -15"}
{"level":30,"time":1771759320249,"pid":18,"hostname":"f86c2c788f60","reqId":"req-8","res":{"statusCode":200},"responseTime":0.685628999559522,"msg":"request completed"}
{"level":30,"time":1771759418130,"pid":18,"hostname":"f86c2c788f60","reqId":"req-9","req":{"method":"GET","url":"/balance/Alice","host":"counter-service:3002","remoteAddress":"172.19.0.4","remotePort":44656},"msg":"incoming request"}
{"level":30,"time":1771759418130,"pid":18,"hostname":"f86c2c788f60","msg":"Returning balance for user Alice: -15"}
{"level":30,"time":1771759418132,"pid":18,"hostname":"f86c2c788f60","reqId":"req-9","res":{"statusCode":200},"responseTime":1.6064679995179176,"msg":"request completed"}
{"level":30,"time":1771759593576,"pid":18,"hostname":"f86c2c788f60","reqId":"req-a","req":{"method":"GET","url":"/balances","host":"counter-service:3002","remoteAddress":"172.19.0.4","remotePort":42202},"msg":"incoming request"}
{"level":30,"time":1771759593576,"pid":18,"hostname":"f86c2c788f60","msg":"Returning all balances, count: 2"}
{"level":30,"time":1771759593577,"pid":18,"hostname":"f86c2c788f60","reqId":"req-a","res":{"statusCode":200},"responseTime":0.5124209998175502,"msg":"request completed"}
```

2. Тестування продуктивності для наведених сценаріїв:

Тестування відбувалося за допомогою модуля load-test-client.

- Сценарій 1: 10 Користувачів, 10000 запитів кожен:

```
-----
RESULTS:
-----
Total requests: 100000
Successful requests: 100000
Failed requests: 0
Total time: 116.66 seconds
Requests per second: 857.22 req/s

Service timing breakdown:
  Logging service: 591.62 s (total time for all workers, 100000 calls, avg: 5.92 ms)
  Counter service: 585.82 s (total time for all workers, 100001 calls, avg: 5.86 ms)
  Logging contribution: 50.25%
  Counter contribution: 49.75%

Final balances:
  user-1: 10000
  user-10: 10000
  user-2: 10000
  user-3: 10000
  user-4: 10000
  user-5: 10000
  user-6: 10000
  user-7: 10000
  user-8: 10000
  user-9: 10000
-----
```

- Сценарій 2: 10 Користувачів, 10000 запитів на спільний рахунок:

```
-----
RESULTS:
-----
Total requests: 100000
Successful requests: 100000
Failed requests: 0
Total time: 119.28 seconds
Requests per second: 838.39 req/s

Service timing breakdown:
  Logging service: 605.90 s (total time for all workers, 100000 calls, avg: 6.06 ms)
  Counter service: 601.73 s (total time for all workers, 100001 calls, avg: 6.02 ms)
  Logging contribution: 50.17%
  Counter contribution: 49.83%

Final balances:
  shared-account: 100000
-----
```

3. Аналіз результатів:

- Найперше спостерігаємо, що обидва сценарії мають дуже схожі результати - час роботи та вклад кожного з сервісів дуже близькі. Це очікувано, адже в нашій архітектурі запити опрацьовуються послідовно, відповідно різниці між роботою з 10-ма окремими рахунками, і 1 спільним, немає. Переважна більшість часу швидше за все витрачена на HTTP запити й Docker, а також роботу з хешмапами / JSON - ці елементи спільні для обох випадків.
- Також бачимо, що затрачений час на **logging** та **counter** сервіси майже однаковий. Це знову ж очікувано, адже основна логіка роботи з обома однакова - обоє отримують запити, додають/змінюють елементи у хешмапі та повертають відповідь. Мінімальну перевагу **counter-service** можемо пояснити тим, що він все ж працює з набагато меншою мапою - 10, або взагалі 1 запис, порівняно з **logging-service**, в якого мапа для кожного логу додає новий.